AFZAL MALIK

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Bachelor of Technology
Electronics Engineering
Aligarh Muslim University (NIRF Rank: 9), Aligarh, U.P

GitHub
LinkedIn

Education

Degree	Institute	Board / University	CGPA/Percentage	Year
B.Tech (ECE)	Zakir Husain College of Engineering and Technology, Aligarh	Aligarh Muslim University	9.4 (Till 6th Sem)	2021-2025
Intermediate	Greenwood Public School, Sitarganj	CBSE	92.8%	2019

Skills

- **Technical Skills**: Cadence Virtuoso, Verilog HDL, Xilinx Vivado, LT Spice Simulation, Electric VLSI, C/C++, Python, MATLAB, Microsoft Office Suite
- **Courses**: Digital & Analog Electronics, Digital System Design, VLSI Design & Technology, Communication System, Control System, Circuit Theory, Semiconductor Device Modelling, Microprocessor

Internship

- **Analog Circuit Design Intern** *June - July 2023*
  Under mentorship of Dr. GS Javed (Analog Design Manager @Intel) Bangalore, India
  - Designed basic building blocks of Analog Integrated Circuits such as: common source amplifier, source follower amplifier, single stage operational amplifier, two stage operational amplifier and filters using gm over Id methodology

Projects

- **Design and FPGA Implementation of Neural Network based Digit Recognition System** *Jan - May 2024*
  B.Tech
  - Tools Used: Xilinx Vivado, VS code | FPGA Board: NEXYS A7 | Language: Verilog HDL & Python
  - Design of software model of Neural Network for Handwritten Digit Recognition system using Python. Hardware realisation of the neural network using Verilog HDL, validating behavioural, Post synthesis & Post Implementation Simulations and then ANN is implemented on FPGA.
- **Design, simulation and layout of Two-Stage Operational Amplifier** *June - July 2023*
  B.Tech **Github**
  - Tools Used: LT Spice and Analog Designer Toolbox (ADT), Electric Binary.
  - Utilized Gm/Id methodology to design and simulate a Two-Stage Operational Amplifier in the 180 nm technology for the specifications: Gain >1000, Gain Bandwidth Product (GBW) > 1GHz, and Phase Margin of 50.
- **Design and Analysis of CMOS Inverter using CMOS 180nm Tcehnology** *Nov 2022*
  B.Tech **Github**
  - Tools Used: LT spice, Electric Binary
  - explored MOSFET models for TSMC180nm, Analyzed strong 0/1 and weak 1/0 logic configurations through simulation. Designed CMOS Inverter, and analysed its voltage transfer characteristics and key design parameters like VOH, VOL, VIH, VIL, and the switching threshold, with its layout using Electric Binary.

Conference Publications

- **A 5GHz Gain-Bandwidth Op-Amp in 180nm technology** *May 2024*
  4th IEEE International Conference on VLSI Systems, Architecture, Technology, and Applications (VLSI SATA 2024) **IEEE Xplore**

Positions of Responsibility

- **Joint Secretary**,Engineering Design & Implementation Club, ZHCET, AMU *Dec. 2022 - till date*
  - Actively contributed to organizing workshops, seminars, and competitions, while promoting collaborative learning within the club

Honors & Awards

- Consistently ranked within the top 3 students of my batch with an overall CPI of 9.403 *2024*
- Awarded scholarship twice from Lead, Ignite & Transform Scholarship (LIT) program based on Merit. *2022, 23*
- Selected as an Analog Circuit Design Intern for Summer Training program *2023*
- Rank 1 in school in Intermediate with 92.8% *2019*